L2-matched active MOLT pairs versus the selected small → hot pair

13 active alternative pairs × 2 sign directions = 26 controls; all injected at layer 22, token 9; attention unfrozen.
Each control’s combined vector is rescaled to intended L2 = 65.3207; right-hand columns show the actual BF16 write.

● −receiver +donor ▼ +receiver −donor ★ Selected pair, 4.5×

Feature names · R = receiver “small”; D = donor “hot”	Same intended intervention norm	Actual applied L2 norm −R +D	Actual applied L2 norm +R −D
SELECTED PAIR · 4.5× **R: L22–T2339 — Small-Size Adjectives and Idiomatic Phrases** **D: L22–T941 — Thermal State and Heat Action Gate**	★ 99.8653%	**65.3121**	—
01 R: L22–T8 — Numbered List and Paragraph Start Detection D: L22–T71 — Proper Noun Subword Boundary Gate	●▼	65.3184	65.3338
02 R: L22–T55 — Think Token Start D: L22–T8 — Numbered List and Paragraph Start Detection	▼●	65.3186	65.3112
03 R: L22–T55 — Think Token Start D: L22–T71 — Proper Noun Subword Boundary Gate	▼●	65.3134	65.3239
04 R: L9–T809 — Small Noun Phrase Gate D: L9–T1348 — Thermal State and Heating Mechanism Gate	▼●	65.3246	65.3181
05 R: L11–T532 — Small-Size Token Gate D: L11–T2210 — Thermal Intensity and Heat Conditions	▼● Best control: 6.55e-07%	65.3256	65.3286
06 R: L11–T485 — Physical Size Adjectives D: L11–T14 — New User Turn Transition	●▼	65.3330	65.3273
07 R: L18–T54 — Final Token Completion Gate D: L18–T51 — User Turn Start	▼●	65.3235	65.3218
08 R: L18–T194 — Sentence-Start Gate Activation D: L18–T1416 — Thermal and Extreme Weather Tokens	●▼	65.3184	65.3248
09 R: L20–T43 — Final Token or Entity Name D: L20–T200 — Code and Technical Token Detection	●▼	65.3360	65.3417
10 R: L20–T399 — Scope-Specifying Phrases Trigger Gate D: L20–T200 — Code and Technical Token Detection	●▼	65.3234	65.3210
11 R: L34–T2338 — Label unavailable in saved dashboard export D: L34–T1550 — Label unavailable in saved dashboard export	●▼	65.3275	65.3224
12 R: L34–T159 — Label unavailable in saved dashboard export D: L34–T401 — Label unavailable in saved dashboard export	▼●	65.3359	65.3358
13 R: L35–T1859 — Label unavailable in saved dashboard export D: L35–T9 — Label unavailable in saved dashboard export	●▼	65.3291	65.3173

10^{-12} 10^{-9} 10^{-6} 10^{-3} 10^{0} 10^{3}

P(cold), % of full vocabulary · logarithmic scale

Actual control norms: 65.3112–65.3417; selected: 65.3121; maximum relative difference: 0.0452%.
C is the full gate × low-rank output × released output scaling. The net pair vector is norm-matched before BF16 rounding.
Other-layer contributions are computed at their source layer, then injected at layer 22. Missing saved labels are marked explicitly.